

SMART INDIA HACKATHON 2025



CODEHEX

- **Problem Statement ID – SIH25070**
- **Problem Statement Title - Secure Data Wiping
for Trustworthy IT Asset Recycling**
- **Theme - Miscellaneous**
- **PS Category - Software**
- **Team ID - SIH25NAWAB14**
- **Team Name(Registered on portal) : CODEHEX**



SYSTEM ARCHITECTURE

SecureWipe Pro



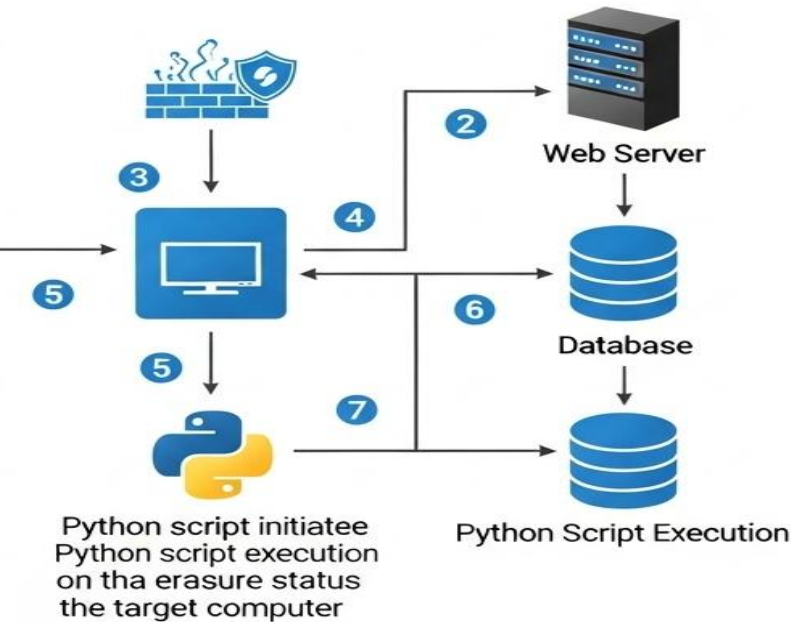
Target Computer



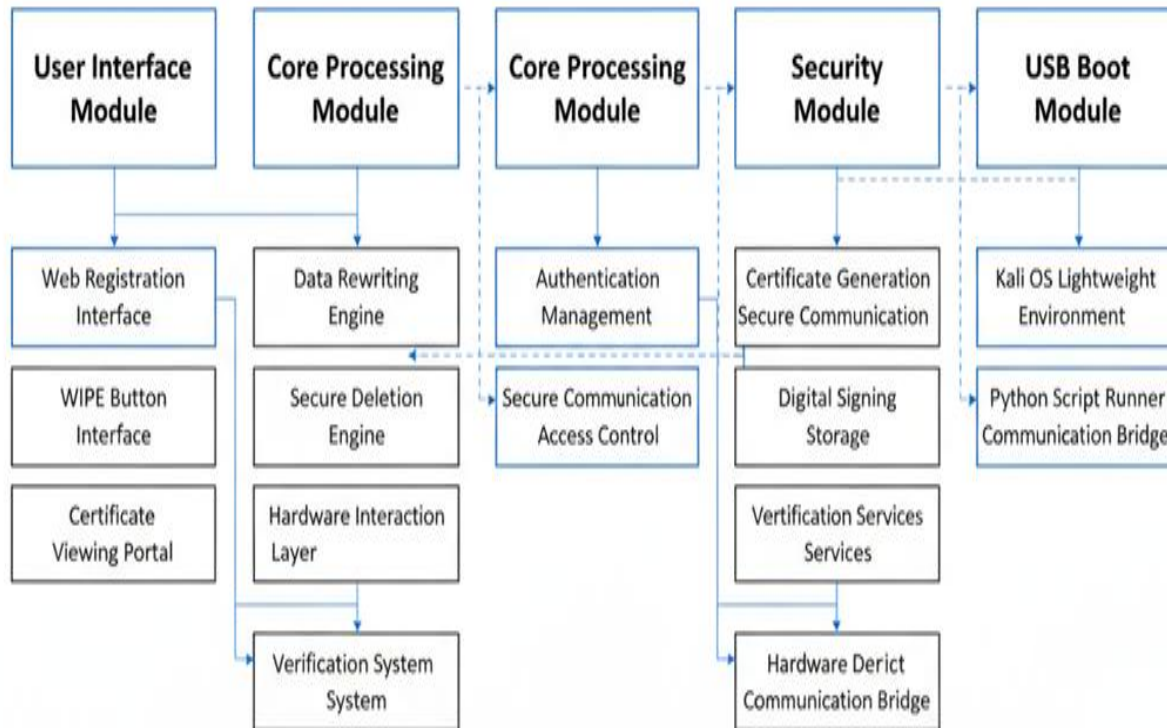
Kali OS USB

LEGEND:

-  Target scripts authentication request
-  Database authentication status
-  Databerver responds
-  HTTPS Connection
-  Encryption

❖ Proposed Solution

- Pre-configured **USB + lightweight OS & Python** ensures **direct hardware access** and bypasses OS restrictions.
- Overwrite + delete** with meaningless patterns for **future-proof security** against recovery.
- Blockchain-based tamper-proof certificate** provides **compliance, transparency & trust**.
- Supports **multiple devices + centralized monitoring**; future upgrades add **encryption & key destruction**.

SecureWipe Pro - General Functional Modules

SecureWipe Pro: Technology Stack



SecureWipe Pro: Project Feasibility & Viability

Feasibility



Technical Feasibility

- Kali OS integration is achievable
- Python script development is straightforward
- Web interface can be built with standard tools



Market Feasibility

- Growing demand for secure data erasure
- IT asset disposal market is expanding
- Compliance requirements drive adoption



Economic Feasibility

- Low development costs
- Subscription model ensures revenue
- Minimal infrastructure requirements

Viability



Scalability

- Cloud-based architecture enables growth
- Modular design allows feature expansion
- Enterprise-ready infrastructure



Security & Trust

- SecureWipe Pro certification builds confidence
- Blockchain verification adds credibility
- Third-party audit compatible



Innovation & Impact

- Novel approach to data security
- Environmental sustainability focus
- Potential industry standard-setter



**TRUSTWORTHY & COMPREHENSIVE
ITAD SOLUTION**

SecureWipe Pro: Benefits & Impact



Social Impact

- Enhanced data privacy protection
- Improved compliance with regulations
- Reduced risk of data breaches
- Greater trust in IT asset disposal



Economic Benefits

- Cost-effective data security solution
- Reduced liability and potential fines
- Value recovery from retired IT assets
- Subscription model for predictable costs



Environmental Impact

- Promotes proper IT asset recycling
- Reduces e-waste through reuse
- Supports sustainable IT reuse
- Supports sustainable IT practices
- Lower carbon footprint



Technological Advancement

- Cutting-cutting data erasure methods
- Blockchain verification technology
- Integration with existing IT systems
- Continuous security improvements

Research & References

IBM Cost of a Data Breach Report 2024 →

<https://www.ibm.com/reports/data-breach> IBM IBM Newsroom

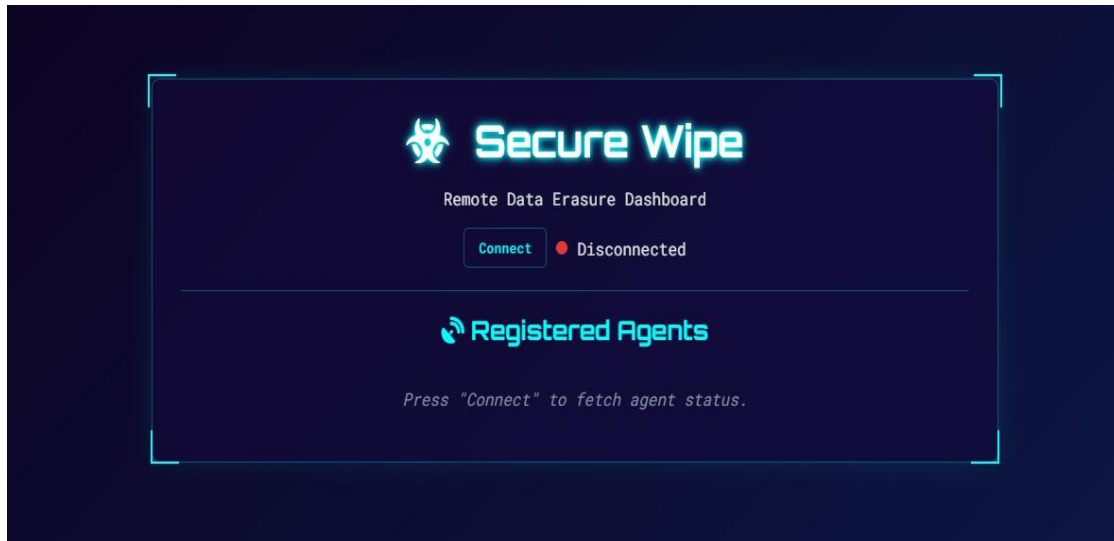
NIST 800-88 Rev.1: Guidelines for Media Sanitization →

<https://csrc.nist.gov/publications/detail/sp/800-88/rev-1/final>

Bootable USB →

https://linuxmint.com/rel_zara.php

Our Website



Comparison with Existing System

Feature / Tool	DBAN	Blancco	SecureWipe Pro 
Wiping Standards	Basic	Advanced	NIST/DoD ready
Certificate Issued	No	Yes	Yes, Auto Digital Proof
Ease of Use	Low	Medium	High (Plug-and-Play)
Cost	Free	Expensive	Low, Reusable USB